

Aryaman Kaprekar

B.Tech Hons. | Cyber Physical Systems | Minor: Computational Intelligence
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SUMMARY

I am driven by curiosity to understand why a system works, where it breaks, and how far it can be pushed. I pursue this through both published research and software deployed on real hardware under real-world constraints. I am interested in pushing AI into critical applications where reliability matters and the stakes are real.

EXPERIENCE

• DeepGrid Semi Pvt Ltd

May 2025 – July 2026

AI Intern → ADAS Engineer → AI Team Lead

- Led the AI team, setting research direction and technical decisions across autonomous driving programs, and represented DeepGrid Semi at the AI Impact Summit and to investors.
- Re-architected the flagship end-to-end autonomous driving model (AD2) for edge hardware using hardware-aware optimization, operator redesign, and FP8 quantization-aware training, cutting compute cost 95% while preserving accuracy.
- Built the full embedded deployment pipeline (PyTorch → ONNX → TensorRT → CUDA → C++) with multi-sensor fusion, raising Jetson Orin Nano throughput $240\times$ ($0.05 \rightarrow 12$ FPS) and cutting model size 86% ($1.1\text{ GB} \rightarrow 150\text{ MB}$).
- Validated on Cityscapes, NuScenes, NuPlan, IDD, and Waymo (open-loop) and CARLA/NavSim (closed-loop), achieving a Drive Score of 86 on Bench2Drive, and co-designed a 28 nm autonomous driving SoC with embedded and RTL teams.

• Advanced Healthcare Lab, MAHE

Dec 2024 - Current

Research Fellow

- Focus areas: lightweight architectures, uncertainty quantification, quantitative explainable AI, and ablation studies for Computer-Aided Diagnostics.
- (Author, Scopus Q1) A multi-level attention CNN-transformer based framework for the detection of brain tumor using regional dual-score explainability. *Nature Scientific Reports (Springer)*, 2026
- (Co-Author, Scopus Q1) TrustNet: A Lightweight Network with Integrated Uncertainty Quantification and Quantitative Explainable AI for Ischemic Stroke Detection in CT Images. *Nature Scientific Reports (Springer)*, 2026
- (Co-Author, Scopus Q2) Automated Identification of Left Ventricular Hypertrophy using Cardiac Ultrasound Imaging: A Systematic Review of AI-Driven Approaches. *Elsevier — Informatics in Medicine Unlocked*, 2025

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech Hons.	Manipal Institute of Technology, Manipal	8.84	2022-2026
Senior Secondary	Maharashtra State Board	82.67%	2022
Secondary	ICSE Board	96.16%	2020

SELECTED PERSONAL PROJECTS

- Road-scenes Segmentation (2025):** Literature review on segmentation; lightweight hybrid GAN model achieving 63% mIoU / 72% mPA on Cityscapes with only 11.3M params.
- Consumer Product Cost Prediction (2025):** Multimodal price prediction fusing ResNet50 + DistilBERT embeddings into an XGBoost regressor; SMAPE of 47.5.
- End-to-end MLOps Pipeline (2025):** Production-grade pipeline with MLFlow, DVC, Docker, and Git-based CI/CD — fully reproducible from raw data to deployed model.
- RL-based Batch Reactor Controller (2025):** RL agent for temperature regulation using Policy Mirror Descent and TD learning for coolant/heater control.
- Generative Models for Reflection Removal (2024):** Autoencoder/GAN pipelines for single-image reflection removal; 89.77% mAP on a custom dataset.

SKILLS

- Programming Languages:** Python, C++, C, Embedded C
- Frameworks & Tools:** PyTorch, ONNX, TensorRT, CUDA, OpenCV, MLFlow, DVC, Docker, Git, Linux, MATLAB, CMake, NumPy, Pandas
- Technical Domains:** Deep Learning, Computer Vision, Model Optimisation, Reinforcement Learning, Edge-AI, Medical Image Analysis, Digital Image Processing, IoT, Computer Architecture, Computer Networks

CERTIFICATIONS AND EXTRA-CURRICULAR

- NPTEL & Coursera:** Top 5% in India across multiple AI/Computer Vision NPTEL courses; Deep Learning Specialization (deeplearning.ai).
- Qualcomm Vision X:** Finalist for the reflection-removal project at IIT-B.
- Workshops:** IoT/ML training at Bolt-IoT; Gesture Robotics at IIT-B.
- Other:** NCC regimental training and social service; Everest Base Camp trek (5,364m); Class Representative.